

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 1

Name: 杨涵

Student ID: 2025531064

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(Story Proof) Define $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ as the number of ways to partition $\{1, 2, \dots, n\}$ into k non-empty subsets, or the number of ways to have n students split up into k groups such that each group has at least one student. For example, $\left\{ \begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right\} = 7$ because we have the following possibilities:

- $\{1\}, \{2, 3, 4\}$ • $\{1, 2\}, \{3, 4\}$
- $\{2\}, \{1, 3, 4\}$ • $\{1, 3\}, \{2, 4\}$
- $\{3\}, \{1, 2, 4\}$ • $\{1, 4\}, \{2, 3\}$
- $\{4\}, \{1, 2, 3\}$

Prove the following identities:

(a)

$$\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}.$$

Hint: I'm either in a group by myself or I'm not.

(b)

$$\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}.$$

Hint: First decide how many people are not going to be in my group.

(a) $\left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\}$ has two situations:
 ① I'm alone in one group $\rightarrow$ the rest n people in $k-1$ groups
 ② I'm part of one group $\rightarrow n$ people in k groups. I have k options

$$\textcircled{1} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} \quad \textcircled{2} = k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}, \text{ so that } \left\{ \begin{smallmatrix} n+1 \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n \\ k-1 \end{smallmatrix} \right\} + k \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$$

(b) $\binom{n}{j}$ means to choose j people from n people.

$\left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}$ to partition $n+1$ people into $k+1$ groups.

Assume j people are not going to be in my group. they should be in k groups. that's $\left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\}$

for a particular number j . j should satisfy $j \geq k$. total number of ways = $\binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\}$

then add all possible j : $\sum_{j=k}^n \binom{n}{j} \left\{ \begin{smallmatrix} j \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right\}$

Problem 2

A *norepeatword* is a sequence of at least one (and possibly all) of the usual 26 letters a, b, c, ..., z, with repetitions not allowed. For example, "course" is a norepeatword, but "statistics" is not. Order matters, e.g., "course" is not the same as "source". A norepeatword is chosen randomly, with all norepeatwords equally likely. Show that the probability that it uses all 26 letters is very close to $1/e$.

$$\text{the number of norepeatwords with } k \text{ letters} = 26 \cdot 25 \cdot (26-k+1) = \frac{26!}{(26-k)!} \quad (\text{order matters})$$

$$\left\{ \begin{array}{l} \text{with 26 letters} = 26! \end{array} \right.$$

$$\text{all norepeatwords} = \sum_{k=1}^{26} \frac{26!}{(26-k)!}, \quad \text{so } p = \frac{26!}{\sum_{k=1}^{26} \frac{26!}{(26-k)!}} = \frac{1}{\sum_{k=1}^{26} \frac{1}{(26-k)!}}. \quad \text{let } j=26-k. \quad \frac{1}{\sum_{j=0}^{25} \frac{1}{j!}}.$$

$$p = \frac{1}{\sum_{j=0}^{25} \frac{1}{j!}}, \quad \text{we know that } e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots = \sum_{j=0}^{\infty} \frac{1}{j!}$$

$$\text{let } e - \sum_{j=0}^{25} \frac{1}{j!} = \frac{1}{26!} + \frac{1}{27!} + \dots \text{ is very small, so that } \sum_{j=0}^{25} \frac{1}{j!} \approx e, \text{ and } p \approx \frac{1}{e}.$$

Problem 3

Given $n \geq 2$ numbers $(a_1, a_2, \dots, a_n)$ with no repetitions, a bootstrap sample is a sequence $(x_1, x_2, \dots, x_n)$ formed from the a_j 's by sampling with replacement with equal probabilities. Bootstrap samples arise in a widely used statistical method known as the bootstrap. For example, if $n = 2$ and $(a_1, a_2) = (3, 1)$, then the possible bootstrap samples are $(3, 3), (3, 1), (1, 3)$, and $(1, 1)$.

- (a) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$?
- (b) How many possible bootstrap samples are there for $(a_1, \dots, a_n)$, if order does not matter (in the sense that it only matters how many times each a_j was chosen, not the order in which they were chosen)?
- (c) One random bootstrap sample is chosen (by sampling from $a_1, \dots, a_n$ with replacement, as described above). Show that not all unordered bootstrap samples (in the sense of (b)) are equally likely. Find an unordered bootstrap sample $\mathbf{b}_1$ that is as likely as possible, and an unordered bootstrap sample $\mathbf{b}_2$ that is as unlikely as possible. Let p_1 be the probability of getting $\mathbf{b}_1$ and p_2 be the probability of getting $\mathbf{b}_2$ (so p_i is the probability of getting the specific unordered bootstrap sample $\mathbf{b}_i$). What is p_1/p_2 ? What is the ratio of the probability of getting an unordered bootstrap sample whose probability is p_1 to the probability of getting an unordered sample whose probability is p_2 ?

(a) $(x_1, x_2, \dots, x_n)$ each place has n options $\Rightarrow n \times n \times \dots = n^n$

(b) order does not matter, so that $a_1, a_2, \dots, a_n$, the total number of all elements being chosen $= n$.

let $l_i =$ how many times a_i is chosen. so $l_1 + l_2 + \dots + l_n = n$, $l_i \geq 0$

* - * - * * - * place $n-1$ bars in $n+n-1$ places. that's $\binom{2n-1}{n-1}$

(c) take $n=3$ as an example, for a_1, a_2, a_3 , all unordered bootstrap samples have 3 situations:

① none repeat: $\{a_1, a_2, a_3\}$ corresponds to $3! = 6$ ordered sequences. $p_1 = \frac{3!}{3^3} = \frac{6}{27} = \frac{2}{9}$

② one element repeats: for example $a_1: \{a_1, a_1, a_2\}, \{a_1, a_1, a_3\}$, each has 3, $p_2 = \frac{3}{27} = \frac{1}{9}$ (x6)

③ 3 repeats: $\{a_1, a_1, a_1\}$. $p_3 = \frac{1}{27}$ (x3)

they are not equally likely.

To find the most likely $\mathbf{b}_1$, each element should be sampled. and $P_1(\max) = \frac{n!}{n^n}$

To find the least likely $\mathbf{b}_2$, only one element is chosen, and $P_2(\min) = \frac{1}{n^n}$

so that $P_1/P_2 = n!$

the probability of $\mathbf{b}_1$ has the $p = p_1$ is $P_1 = p_1$, cause only one sequence $\{a_1, \dots, a_n\}$ satisfies.

The probability of $\mathbf{b}_1$ has the $p = p_2$ is $P_2 = n \cdot p_2$, cause $a_1, a_2, \dots, a_n$, each can repeat n times. $\rightarrow n$ sequences

so that $\frac{P_1}{P_2} = \frac{1}{n} \cdot n! = (n-1)!$

Problem 4

In the birthday problem, we assumed that all 365 days of the year are equally likely (and excluded February 29). In reality, some days are slightly more likely as birthdays than others. For example, scientists have long struggled to understand why more babies are born 9 months after a holiday. Let $\mathbf{p} = (p_1, p_2, \dots, p_{365})$ be the vector of birthday probabilities, with p_j the probability of being born on the j th day of the year (February 29 is still excluded, with no offense intended to Leap Dayers). The k th elementary symmetric polynomial in the variables $x_1, \dots, x_n$ is defined by

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} x_{j_1} \dots x_{j_k}.$$

This just says to add up all of the $\binom{n}{k}$ terms we can get by choosing and multiplying k of the variables.

For example, $e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$, $e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$, and $e_3(x_1, x_2, x_3) = x_1x_2x_3$. Now let $k \geq 2$ be the number of people.

- Find a simple expression for the probability that there is at least one birthday match, in terms of $\mathbf{p}$ and an elementary symmetric polynomial.
- Explain intuitively why it makes sense that P (at least one birthday match) is minimized when $p_j = \frac{1}{365}$ for all j , by considering simple and extreme cases.
- The famous arithmetic mean-geometric mean inequality says that for $x, y \geq 0$

$$\frac{x+y}{2} \geq \sqrt{xy}.$$

This inequality follows from adding $4xy$ to both sides of $x^2 - 2xy + y^2 = (x-y)^2 \geq 0$. Define $\mathbf{r} = (r_1, \dots, r_{365})$ by $r_1 = r_2 = (p_1 + p_2)/2$, $r_j = p_j$ for $3 \leq j \leq 365$. Using the arithmetic mean-geometric mean bound and the fact, which you should verify, that

$$e_k(x_1, \dots, x_n) = x_1x_2e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2)e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$$

show that

$$P(\text{at least one birthday match} \mid \mathbf{p}) \geq P(\text{at least one birthday match} \mid \mathbf{r})$$

with strict inequality if $\mathbf{p} \neq \mathbf{r}$, where the given $\mathbf{r}$ notation means that the birthday probabilities are given by $\mathbf{r}$. Using this, show that the value of $\mathbf{p}$ that minimizes the probability of at least one birthday match is given by $p_j = \frac{1}{365}$ for all j .

(a) The probability of k people all have different birthday. $= k! \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq n} p_{j_1} \dots p_{j_k} = k! e_k(\mathbf{p})$
 so the probability that there is at least one birthday match. $p = 1 - k! e_k(\mathbf{p})$

(b) if there are m days available and only two people. $P(\text{match}) = \sum_{j=1}^m p_j^2$.
 we know that $\sum_{j=1}^m (p_j - \frac{1}{m})^2 = \sum_{j=1}^m p_j^2 + m \cdot \frac{1}{m^2} - \frac{2}{m} = \sum_{j=1}^m p_j^2 - \frac{1}{m} \geq 0 \Rightarrow \sum_{j=1}^m p_j^2 \geq \frac{1}{m}$,
 $\sum_{j=1}^m p_j^2 = \frac{1}{m}$ holds true only when $p_1 = p_2 = \dots = p_j = \frac{1}{m}$.
 so that intuitively, P is minimized when $p_j = \frac{1}{365}$.

(4) ① verify the equation first, for $e_k(x_1, \dots, x_n)$ we have three situations:

1. x_1, x_2 are in, that's $x_1 x_2 e_{k-2}(x_3, \dots, x_n)$
2. x_1 or x_2 is in, that's $(x_1 + x_2) e_{k-1}(x_3, \dots, x_n)$
3. none is in, that's $e_k(x_3, \dots, x_n)$

$$\text{so that } e_k(x_1, \dots, x_n) = x_1 x_2 e_{k-2}(x_3, \dots, x_n) + (x_1 + x_2) e_{k-1}(x_3, \dots, x_n) + e_k(x_3, \dots, x_n)$$

let $r = (r_1, \dots, r_{365})$, which $r_1 = r_2 = \frac{p_1 + p_2}{2}$, $r_j = p_j$ for $3 \leq j \leq 365$.

$$P(\text{match}|p) = 1 - k! e_k(p), \quad P(\text{match}|r) = 1 - k! e_k(r)$$

$$P(\text{match}|p) - P(\text{match}|r) = k! [e_k(r) - e_k(p)]$$

$$e_k(r) - e_k(p) = r_1 r_2 e_{k-2}(r_3, \dots, r_{365}) + (r_1 + r_2) e_{k-1}(r_3, \dots, r_{365}) + e_k(r_3, \dots, r_{365}) - [p_1 p_2 e_{k-2}(p_3, \dots, p_{365}) + (p_1 + p_2) e_{k-1}(p_3, \dots, p_{365}) + e_k(p_3, \dots, p_{365})]$$

since $p_j = r_j$ for $3 \leq j \leq 365$, let $A = (p_3, \dots, p_{365})$

$$e_k(r) - e_k(p) = e_{k-2}(A) \cdot (r_1 r_2 - p_1 p_2) + e_{k-1}(A) \cdot (r_1 + r_2 - p_1 - p_2)$$

$$r_1 = r_2 = \frac{p_1 + p_2}{2}, \quad r_1 + r_2 - p_1 - p_2 = 0, \quad e_k(r) - e_k(p) = 0,$$

$$r_1 r_2 - p_1 p_2 = \frac{1}{4} (p_1^2 + p_2^2 + 2p_1 p_2) - p_1 p_2 = \frac{1}{4} (p_1^2 + p_2^2 - 2p_1 p_2) = \frac{1}{4} (p_1 - p_2)^2 \geq 0, \text{ so that } e_k(r) - e_k(p) \geq 0.$$

therefore, $P(\text{match}|p) \geq P(\text{match}|r)$. if $p \neq r$, it is strictly unequal.

Given an arbitrary vector $p = (p_1, \dots, p_{365})$, $p_1 + \dots + p_{365} = 1$.

to minimise the probability, take $p_1' = p_2' = \frac{p_1 + p_2}{2}$, $p_2'' = p_3'' = \frac{p_2' + p_3'}{2} \dots$

after a large number of iterations, the final minimum p will result in $p_j = \frac{1}{365}$ for all j .